UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 007 · Taylor's and Maclaurin's expansion · compiled 14 October 2026

Contents

1. Short notes
2. Keyword index
3. Rapid recall
4. Last 10 minutes

1 Short notes

1 · Why Taylor's and Maclaurin's series?

Series turn complicated functions into polynomials, which can be evaluated, integrated and used in limits: $\int_0^1 \sin(x^2)dx \approx \frac{1}{3} - \frac{1}{42} + \frac{1}{1320} = 0.3103$; $\lim \frac{\sin x - x}{x^3} = -\frac{1}{6}$. Applications: integrals, asymptotics, differential equations, Lorentz factor, $\sin\theta \approx \theta$, optimization, visualization.

2 · Taylor's and Maclaurin's series in one variable

Taylor about a: $f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \ldots$; Maclaurin about 0. Needs finite derivatives at the point. Standard: e^x, $\log(1+x)$, $(1+x)^n$, $\sin$, $\cos$, $\sinh$, $\cosh$, $\tan^{-1}$, $\sin^{-1}$, $\cos^{-1}$; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.

3 · Taylor's expansion of a function of two variables

About (a,b) with $h = x-a$, $k = y-b$: $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$, all at (a,b). ESA: $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$ gives $\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3$.

4 · Maclaurin's expansion of a function of two variables

About $(0,0)$: $f + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \ldots$ ESA: $\log(1+x-y) = (x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3}$; $e^{ax}\sin by = by + abxy + \frac{a^2 b}{2}x^2 y - \frac{b^3}{6}y^3$; $\sin(x+2y) = (x+2y) - \frac{(x+2y)^3}{6}$.

2 Keyword index

18 terms; the number in brackets is the topic.

Approximate value (3). Evaluate the truncated series at a nearby point.

Definite integral by series (1). Integrate the series term by term.

In powers of (x − a) (3). About the point (a, b).

Lorentz factor (1). Approximated by its Maclaurin series at low speed.

Maclaurin's series (2). Taylor's series about the origin.

Maclaurin's series in two variables (4). E4.1 about $(0, 0)$.

Mixed partials (3). $f_{xy} = f_{yx}$; give the coefficient 2.

One-combination shortcut (4). $f(x + 2y)$: use the one-variable series.

Pendulum approximation (1). $\sin\theta \approx \theta$ (one term).

Polynomial approximation (1). Truncating the series after a few terms.

Radius of validity (2). $|x| < 1$ for $\log(1 + x)$, $(1 + x)^n$, $\frac{1}{1-x}$.

Standard series (2). e^x, $\log(1 + x)$, $\sin x$, ... (slide 007·5).

Tangent line (2). First two terms of Taylor's series.

Tangent plane (3). First-degree Taylor polynomial in two variables.

Taylor's series (2). Polynomial form of f about a point.

Taylor's series in two variables (3). E3.1 about (a, b).

Taylor's theorem (1). Basis of L'Hôpital's rule and the mean value theorems.

Third-degree terms (3). $\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$.

3 Rapid recall

1	Taylor's series of $f(x)$ about $x=a$?	$f(a)+(x-a)f'(a)+\frac{(x-a)^2}{2!}f''(a)+\dots$
2	Maclaurin's series is Taylor's series about?	The origin.
3	Condition for Maclaurin's series?	f and all derivatives finite at 0 (series convergent).
4	e^x?	$1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\dots$
5	$\log(1+x)$, and where?	$x-\frac{x^2}{2}+\frac{x^3}{3}-\dots$, $\lvert x\rvert<1$.
6	$\sin x$?	$x-\frac{x^3}{3!}+\frac{x^5}{5!}-\dots$
7	$\cos x$?	$1-\frac{x^2}{2!}+\frac{x^4}{4!}-\dots$
8	$\tan^{-1}x$?	$x-\frac{x^3}{3}+\frac{x^5}{5}-\dots$
9	Does $1+2+4+\dots$ equal -1?	No, it diverges; $\frac{1}{1-x}$ needs $\lvert x\rvert<1$.
10	$\int_0^1 \sin(x^2)dx$?	$\frac{1}{3}-\frac{1}{42}+\frac{1}{1320}-\dots\approx 0.3103$.
11	$\lim \frac{\sin x-x}{x^3}$?	$-\frac{1}{6}$.
12	Two-variable second-order bracket?	$\frac{1}{2!}\{h^2f_{xx}+2hkf_{xy}+k^2f_{yy}\}$.
13	Third-order bracket?	$\frac{1}{3!}\{h^3f_{xxx}+3h^2kf_{xxy}+3hk^2f_{xyy}+k^3f_{yyy}\}$.
14	Why the 2 in $2hkf_{xy}$?	$f_{xy}=f_{yx}$.
15	First-degree Taylor polynomial is the ...?	Tangent plane.
16	$\sin x\sin y$ about $(\frac{\pi}{4},\frac{\pi}{4})$, second order?	$\frac{1}{2}+\frac{1}{2}(h+k)-\frac{1}{4}(h-k)^2$.
17	Its third-degree term?	$-\frac{1}{12}(h+k)^3$.
18	$\log(1+x-y)$ to third degree?	$(x-y)-\frac{(x-y)^2}{2}+\frac{(x-y)^3}{3}$.
19	$e^{ax}\sin by$ to third degree?	$by+abxy+\frac{a^2b}{2}x^2y-\frac{b^3}{6}y^3$.
20	$\sin(x+2y)$ to third degree?	$(x+2y)-\frac{(x+2y)^3}{6}$.
21	$e^x\cos y$ about $(1,\frac{\pi}{4})$, second degree?	$\frac{e}{\sqrt{2}}[1+h-k+\frac{1}{2}(h^2-2hk-k^2)]$.
22	$\tan^{-1}\frac{0.9}{1.1}$ by Taylor?	≈ 0.6854.

Mnemonic (study aid). *"Binomial brackets."* Order n is $\frac{1}{n!}(h\partial_x+k\partial_y)^n f$: coefficients 1, 2, 1 at second order and 1, 3, 3, 1 at third.

4 Last 10 minutes before the exam

Top facts

1. Taylor about a: $f(a)+(x-a)f'(a)+\frac{(x-a)^2}{2!}f''(a)+\ldots$; Maclaurin: about the origin.
2. Needs finite derivatives at the point; $\log(1+x)$, $(1+x)^n$, $\frac{1}{1-x}$ only for $|x|<1$.
3. Two variables: $f+\{hf_x+kf_y\}+\frac{1}{2!}\{h^2f_{xx}+2hkf_{xy}+k^2f_{yy}\}+\ldots$, $h=x-a$, $k=y-b$.
4. Third order: $\frac{1}{3!}\{h^3f_{xxx}+3h^2kf_{xxy}+3hk^2f_{xyy}+k^3f_{yyy}\}$ (asked in 4 of 6 ESA questions).
5. Method: derivative table, evaluate at the point, substitute order by order.
6. One combination ($x+2y$, $x-y$): use the one-variable series.
7. ESA: $\sin x\sin y$ about $(\frac{\pi}{4},\frac{\pi}{4})$; $e^x\cos y$ about $(1,\frac{\pi}{4})$; $\log(1+x-y)$; $e^{ax}\sin by$; $\sin(x+2y)$.
8. First degree = tangent plane; more terms follow the surface further.

Top formulas

E2.1 · Taylor's series, one variable — ESA ★

$$f(x)=f(a)+(x-a)f'(a)+\frac{(x-a)^2}{2!}f''(a)+\cdots+\frac{(x-a)^n}{n!}f^{(n)}(a)+\ldots$$

GIVES A polynomial form of f near $x=a$.

APPLIES f has derivatives of every order at a and the series converges to f.

NOTE Slide 007·4 writes $f^n(a)$ for the n-th derivative $f^{(n)}(a)$.

S007·4 · T007 00:04:56

E2.2 · Maclaurin's series, one variable — ESA ★

$$f(x)=f(0)+xf'(0)+\frac{x^2}{2!}f''(0)+\cdots+\frac{x^n}{n!}f^{(n)}(0)+\ldots$$

GIVES Taylor's series about the origin.

APPLIES f and all its derivatives finite at 0 (not, e.g., $1/x$ or $\log x$).

NOTE "About a point $x=a$: Taylor; about the origin: Maclaurin."

S007·4 · T007 00:11:17

E3.1 · Taylor's series, two variables (to second order) — ESA ★★★

$$f(x,y)=f+\frac{1}{1!}\{hf_x+kf_y\}+\frac{1}{2!}\{h^2f_{xx}+2hkf_{xy}+k^2f_{yy}\}+\ldots,\quad h=x-a,\ k=y-b$$

GIVES A polynomial form of f near (a,b).

APPLIES All derivatives evaluated at (a,b), all finite there.

NOTE h, k are shorthand added for clarity; the slide writes $(x-a)$, $(y-b)$.

S007·9 · T007 00:18:07 · T007 00:18:45

E3.2 · Third-order term

ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES The terms needed for "up to third degree".

APPLIES Add to E3.1 (or E4.1 with $h = x$, $k = y$).

NOTE Not on the slides; asked in four of six ESA Taylor questions.

ADDED FOR CLARITY ESA Dec-23 1b · ESA Jan-24 s1 1b · ESA Jan-24 s2 1b · ESA Jun-24 1c

E4.1 · Maclaurin's series, two variables

ESA ★★★

$$f(x,y) = f(0,0) + \frac{1}{1!}\{x f_x + y f_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}\} + \dots$$

GIVES A polynomial form of f near the origin.

APPLIES Derivatives at $(0,0)$, all finite there.

NOTE Same as E3.1 with $a = b = 0$; add E3.2 with $h = x$, $k = y$ for third degree.

S007·10 · T007 00:19:38

E2.5 · Trigonometric and hyperbolic

ESA ★★

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots, \qquad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

GIVES Odd and even standard series.

APPLIES All real x.

NOTE Slide 007·5 prints $\cos x = 1 - \frac{x^2}{2} + \dots$ ($2! = 2$).

S007·5